DMAIC Project Charter

Capacity Fade Characterization and Quality Control of Lithium-Ion Cells using NASA PCoE Cycling Data

Project Lead	Bhargava Sriram P	**Sponsor**	Self-directed
Start	October 2026	**Target Close**	31 January 2027

Business Case

Battery cell manufacturers lose yield and reliability when capacity fade characteristics vary across cells. A reproducible Six Sigma analysis on NASA PCoE reference data establishes the measurement and control baseline needed for high-volume cell production.

Problem Statement

NASA PCoE cells B0005, B0006, B0007, B0018 show capacity fade from approximately 1.85 Ah to below 1.4 Ah (80 percent End-of-Life threshold) over 100 to 170 cycles. Cell-to-cell variation in this fade behavior is not currently quantified under Statistical Process Control.

Goal Statement

Quantify capacity fade behavior, establish Statistical Process Control limits, perform Cp and Cpk capability study, and build a Weibull reliability model. All deliverables reproducible in both Python and JMP by 31 January 2027.

Scope — In

- NASA PCoE cells B0005, B0006, B0007, B0018
- Python pipeline (pandas, NumPy, SciPy, matplotlib)
- JMP replication of all statistical outputs
- DMAIC phases Define, Measure, Analyze, Improve, Control

Scope — Out

- Sthyr Energy proprietary zinc-air data (under NDA)
- Proprietary industry chemistries (iron-air, nickel-metal hydride, solid-state)
- Physical cell cycling or lab experiments

Deliverables

- DMAIC Project Charter, CTQ Tree, SIPOC (Define)
- Measurement System Analysis report (Measure)
- SPC control charts: X-bar / R and I-MR with Western Electric rules (Analyze)
- Pareto, Fishbone, ANOVA on fade drivers (Analyze)
- 2-squared Design of Experiments on temperature and cutoff voltage (Improve)
- Cp / Cpk capability study and Weibull reliability model (Control)
- Final report, slide deck, and public GitHub repository

Project Team

Name	Role
Bhargava Sriram P	Project Lead, Analyst, Author